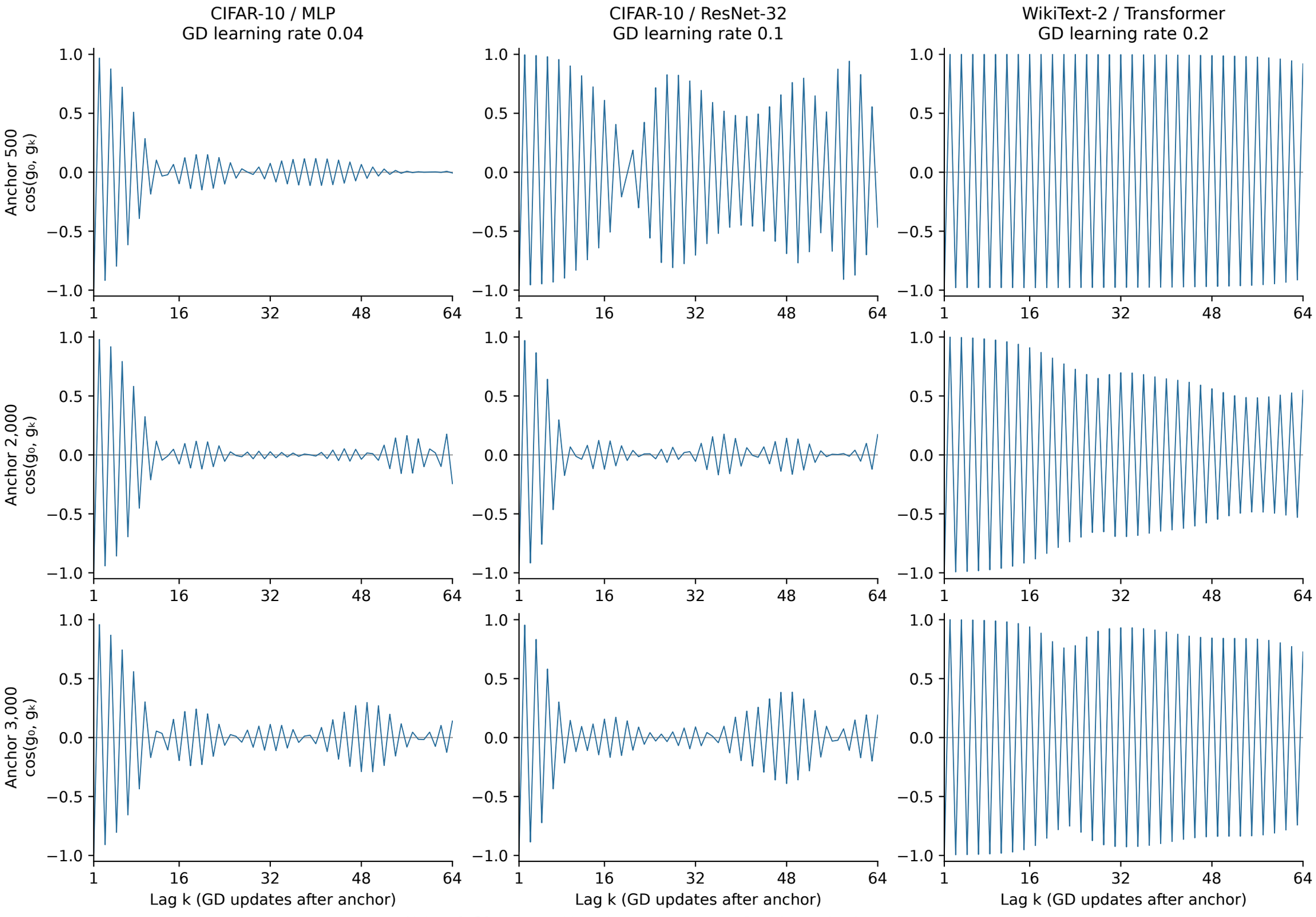

Anchored raw-gradient cosine at the largest GD learning rate
CIFAR-10 / MLP
GD learning rate 0.04
CIFAR-10 / ResNet-32
GD learning rate 0.1
WikiText-2 / Transformer
GD learning rate 0.2
Anchor 500
cos(g₀, gₖ)
Anchor 2,000
cos(g₀, gₖ)
Anchor 3,000
cos(g₀, gₖ)
1.0
0.5
0.0
−0.5
−1.0
1
16
32
48
64
Lag k (GD updates after anchor)
All lags 1–64 connected in order; each row starts from its retained anchor.